8.3 Analyzing Geometric Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.3 · Student sheet

I can…

- tell whether a sequence is geometric and write its rule.
- write a rule from one term and the common ratio, or from two terms.
- find the sum of a finite geometric series.

Vocabulary · fill in the missing word

Geometric sequence — the ratio of _____ terms is constant

Common ratio r — the _____ ratio

Rule — a_n = _____

Sum of a finite geometric series — S_n = _____, $r \neq 1$

Part 1 · Spreading the news

On day 1 you tell 3 friends a secret. Each day, everyone who heard it the day before tells 3 new people.

Day	1	2	3	4	5	6
New people						
Total who know						

Rule for the new people on day n: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Geometric sequence: Geometric: the _____ of consecutive terms is constant (r).

Example 1

Tell whether each sequence is geometric.

a. 5, 10, 20, 40, … b. 1, 4, 9, 16, … c. 96, −48, 24, −12, …

Rule: Rule: a_n = _____

Example 2

Write a rule for the nth term of 4, 20, 100, 500, … Then find a_7.

Example 3

Two terms of a geometric sequence are $a_2 = 12$ and $a_5 = -768$. Write a rule for the nth term.

Sum of a finite geometric series: Sum: S_n = _____

Example 4

Find Σ (i = 1 to 8) $4(3)^{i-1}$.

Example 5

A scholarship fund gives $2000 the first year and 5% more each year after that. How much does it give in all over 10 years?

Part 3 · Partner practice

Partner 1

One term of a geometric sequence is $a_4 = 12$, and r = 2. Write a rule for the nth term.

Partner 2

Find the sum of the first 6 terms of 1, −2, 4, −8, …